

MUHAMMET ALİ GEDİK

COMPUTER ENGINEERING STUDENT

Istanbul, Turkey • +90 (552) 572 4758 • ali.101gedik@gmail.com • github.com/gedik25

SUMMARY

Third-year Computer Engineering student with a strong foundation in software development languages including C, Java, and Python. Passionate about Artificial Intelligence and eager to learn new technologies. Possesses fundamental knowledge in networking and database management. Looking for a challenging internship/role to leverage academic experience and contribute to dynamic engineering teams.

EDUCATION

B.Sc. in Computer Engineering

2023 - Present

Marmara University, Istanbul, Turkey

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (Java), Database Management Systems, Operating Systems, Computer Networks.

High School Diploma (Science & Mathematics Focus)

2019 - 2023

Erzurum Necmettin Erbakan Science High School, Erzurum, Turkey

Completed rigorous curriculum focused on Advanced Mathematics, Physics, and Natural Sciences.

PROJECTS

PAC-MAN (16-bit DOS Assembly)

2025

Technologies: x86 Assembly (16-bit), MS-DOS, DOSBox, COM Executable

- Developed a fully functional clone of the classic PAC-MAN game using 16-bit x86 Assembly language for MS-DOS.
- Optimized memory management and code efficiency to achieve an ultra-compact executable size of approximately ~1 KB.
- Implemented core game mechanics, including character movement, input handling, and collision detection, at the register level without high-level abstractions.
- Ensured cross-platform compatibility on modern systems via DOSBox emulation.

Library Management System

2025

Technologies: Java 21, JavaFX, Maven, MVC Architecture, File I/O

- Developed a desktop application using **JavaFX** and **MVC architecture** to manage library inventories and user transactions efficiently.
- Engineered the backend using advanced **OOP principles** (Inheritance, Abstraction, Encapsulation) and implemented custom **exception handling** logic.

- Designed role-based access control (Admin & Visitor) and a file-based data persistence system using **Java File I/O**.

Hospital Network Infrastructure Design

2025

Technologies: Cisco Packet Tracer, VLAN, DHCP, Static Routing

- Designed a simulated network topology for a hospital environment using **Cisco Packet Tracer**.
- Implemented **VLANs** to segment network traffic between different departments (e.g., Administration, Medical Staff) for better organization.
- Configured **DHCP servers** for dynamic IP allocation and established **Static Routing** to ensure connectivity between different subnets.

Movie & TV Show Tracking Application

2025

Technologies: Python, Tkinter (GUI), JSON

- Developed a desktop application using Python and Tkinter to track and manage personal movie/TV show watchlists with a user-friendly GUI.
- Implemented a persistent storage system using JSON to ensure data integrity for user entries.
- Designed interactive features including a 0-5 star rating slider and dynamic watch status toggles (Watched/Unwatched).
- Engineered full CRUD (Create, Read, Update, Delete) capabilities allowing users to manage categories, genres, and content details efficiently.

SKILLS

Programming Languages: Python, C, Java, Assembly (x86), SQL.

Libraries & Frameworks: PyTorch, TensorFlow, JavaFX, Tkinter, Pandas, NumPy.

Tools & Platforms: Git, GitHub, Linux (Ubuntu/Fedora), VS Code, Cisco Packet Tracer.

Concepts: OOP, MVC Architecture, Data Structures, Algorithms.

LANGUAGES

Turkish • English (Professional Working Proficiency)

WEB & SOCIAL

github.com/gedik25 • linkedin.com/in/muhammetaligedik